Topological analysis of Cold and Self-Interacting Dark Matter models

LIU oral defense

M2 IRT at Observatory of Paris (PSL)

LIU at IAP

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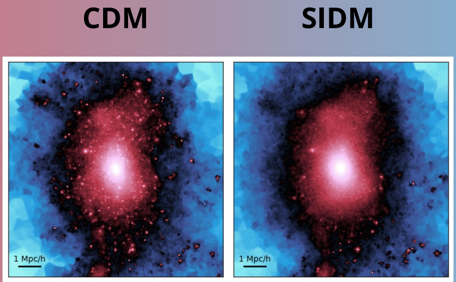
supervised by C. Laigle (IAP) and P. Boldrini (LIRA)

Academic Year 2025/2026



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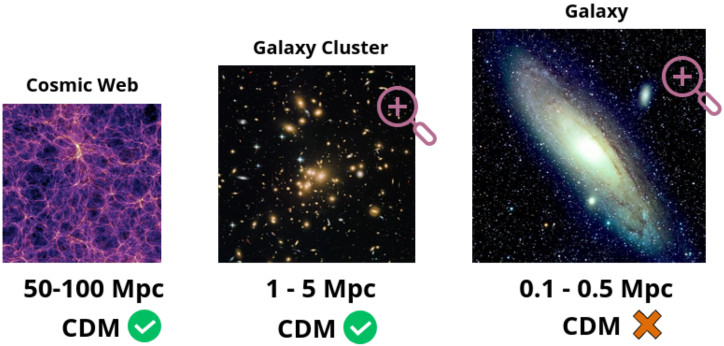
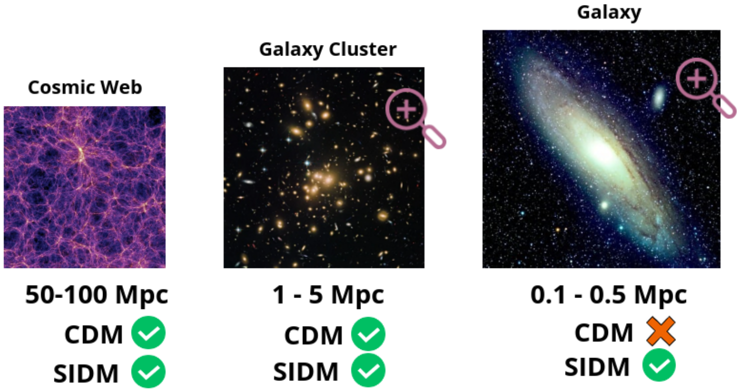
Standard cosmological model : ΛCDM , 27% of the universe made of **Cold Dark Matter (CDM)**.



Scientific context

Standard cosmological model : ΛCDM , 27% of the universe made of **Cold Dark Matter (CDM)**.

$\rightarrow$ Alternative model: **Self-Interacting Dark Matter (SIDM)**, introduces **self-collisions**, change the dynamics at **galactic scales**.



Objective of this work

Problem: Observations are difficult to interpret because of **complex baryonic effects**.

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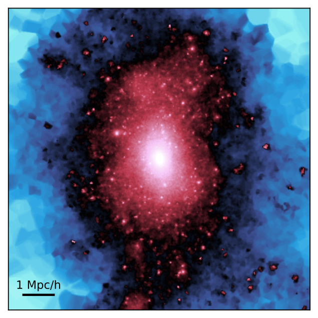
Objective: Use cosmological information to distinguish CDM and SIDM models in cosmological simulations.

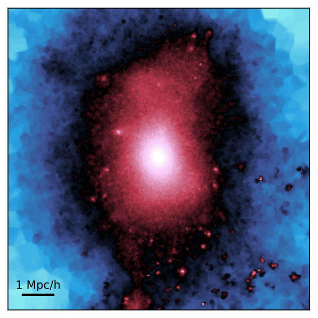
CDM vs SIDM: what is the difference?

Cold



- **Collisionless** particles, interact only gravitationally.
- Low kinetic energy $\Rightarrow$ particles can be **trapped** in small-scale structures.

Self-Interacting



- **Collisions** quantified by the **Cross-section** σ_0.
- Self-collisions $\Rightarrow$ speed increase $\Rightarrow$ elongated orbits $\Rightarrow$ **smoother density**.

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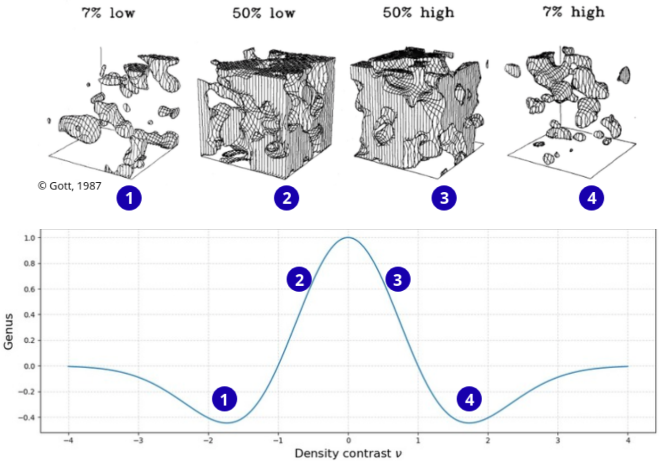
Genus statistics

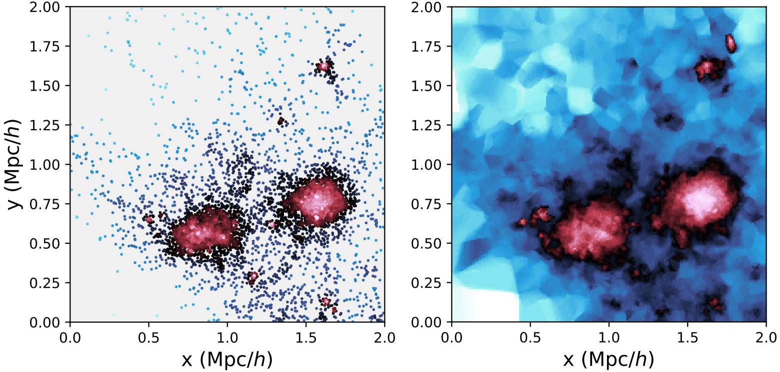
$$\boxed{\text{Genus} = \text{Nb. of holes} - \text{Nb. of isolated regions}}$$



How to compute the genus and clump abundance?

I. Interpolate the 3D density field from the simulation

From 576^3 particle positions in a $(48\,\mathrm{Mpc}/h)^3$ box to a 1000^3 grid via Delaunay Tessellation.



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Gaussian filter of width R_G to **probe structures at different scales**.
Normalize the density field: $\nu = \frac{\rho - \bar{\rho}}{\sigma}$.

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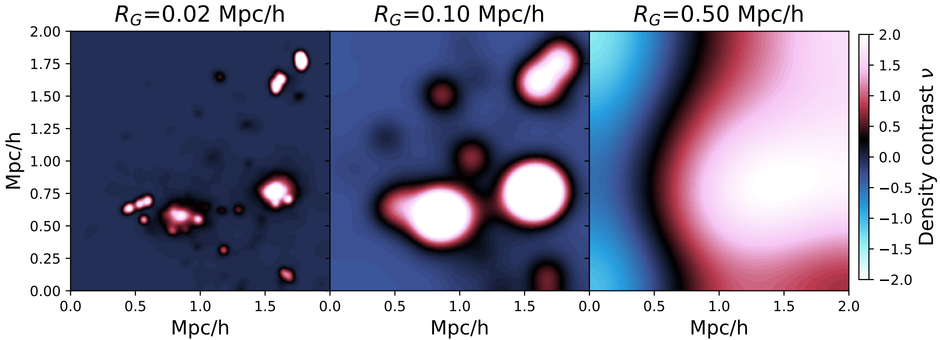
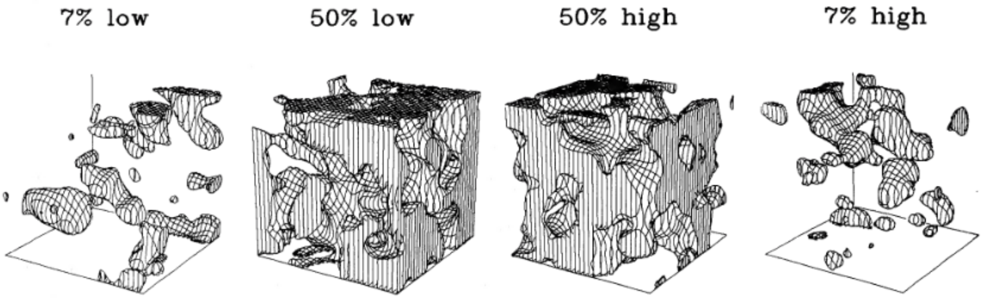


How to compute the genus and clump abundance?

III. Define isodensity surfaces and compute the genus

Genus = Nb. holes − Nb. isolated regions.



Genus < 0 Genus > 0 Genus < 0

Isodensity surface at a given threshold. Credit: Gott, 1987.

How to compute the genus and clump abundance?

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From 576^3 particle positions in a $(48\,\mathrm{Mpc}/h)^3$ box to a 1000^3 grid via Delaunay Tessellation.

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III. Define isodensity surfaces and compute the genus

Genus $=$ Nb. holes $-$ Nb. isolated regions.

IV. Vary the density contrast and compute the genus curve

Divide the box in 8 sub-boxes and compute the genus separately.

Application to the simulation snapshots

Generalize the method to many selected **dense regions**.

1. **Select** and **zoom in** on a sample of DM halos.
2. **Compute** the genus.
3. **Repeat** the computation for **CDM and SIDM** snapshots.

Note: Total computation time for 24 halos: 15 CPU hours.

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Comparison of the genus curves

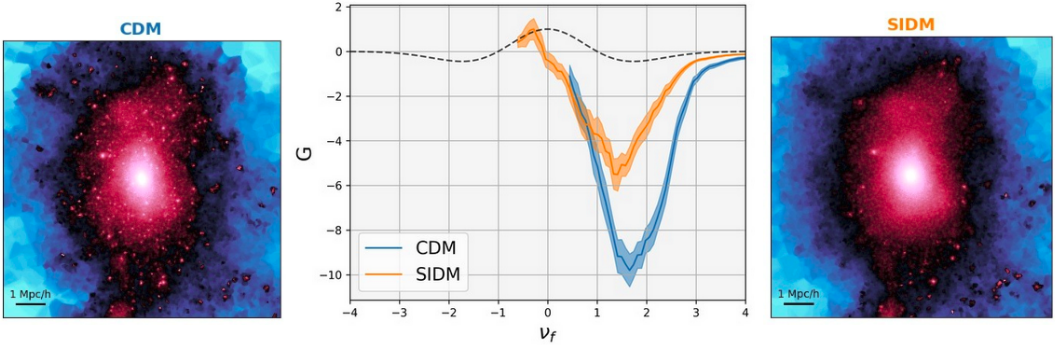

Figure: Halo of mass $M_{\mathrm{halo}} \approx 10^{14}\,\mathrm{M}_{\odot}/h$ at $z = 0$, smoothed at $R_G = 0.01\,\mathrm{Mpc}/h$. CDM.

Clump abundance

3 Results: comparison Cold/Self-Interacting Dark Matter

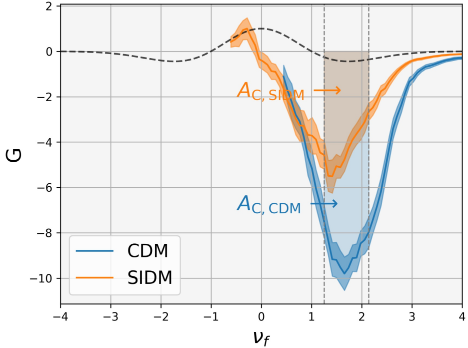

Figure: Halo of mass $M_{\text{halo}} \approx 10^{14}\,\mathrm{M}_{\odot}/h$ at $z = 0$, smoothed at $R_G = 0.01\,\mathrm{Mpc}/h$. SIDM with $\sigma_0 = 1\mathrm{cm}^2/\mathrm{g}$.

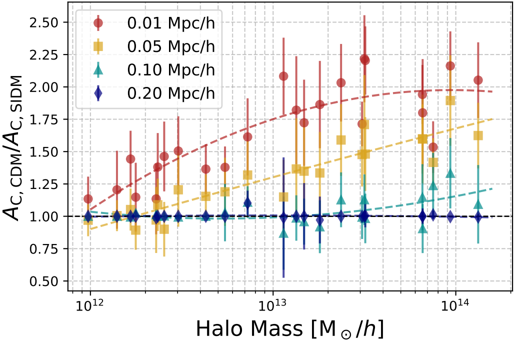

Figure: 24 halos at redshift $z = 0$.

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4 Conclusion

Conclusion and summary

- **New contribution:** Use of **topological analysis** (genus) to compare CDM and SIDM.

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- **Main results:** CDM is **clumpier** than SIDM for smoothing scales $R_G \lesssim 0.1\,\mathrm{Mpc}$ and **identical** above.
  $\rightarrow$ Largest differences found at **low redshift** ($z \lesssim 1$) around **massive halos** ($M_{\mathrm{halo}} \gtrsim 10^{12}\,\mathrm{M}_{\odot}$).

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- **Main results:** CDM is **clumpier** than SIDM for smoothing scales $R_G \lesssim 0.1\,\mathrm{Mpc}$ and **identical** above.
  $\rightarrow$ Largest differences found at **low redshift** ($z \lesssim 1$) around **massive halos** ($M_{\mathrm{halo}} \gtrsim 10^{12}\,\mathrm{M}_{\odot}$).
- **Connect to observables:** applied the analysis to density field calculated from the halo catalog.
- **Future prospects:**
  - improve the the analysis on halo density field and include observational effects $\rightarrow$ publication in preparation.
  - apply the method to other dark matter models (Fuzzy DM, Warm DM).

Thank you for listening!
Any questions?

Comparison of the genus curves (complete 48 Mpc/h box)

5 Appendix A

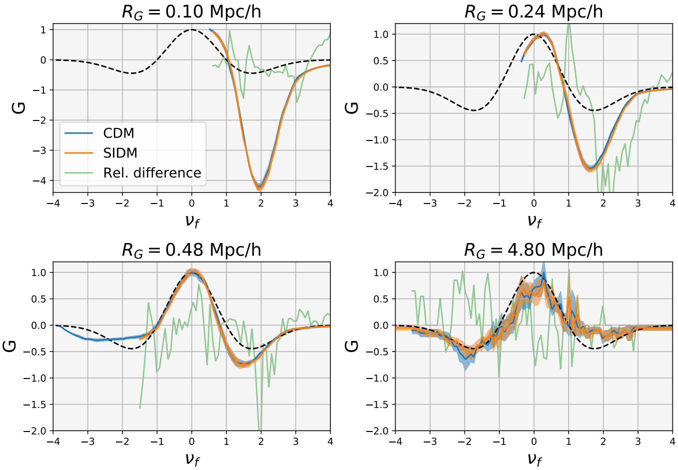

Figure: Genus computed on the halo density field. SIDM with $\sigma_0 = 1\mathrm{cm}^2/\mathrm{g}$.

Comparison of the genus curves (halo density field)

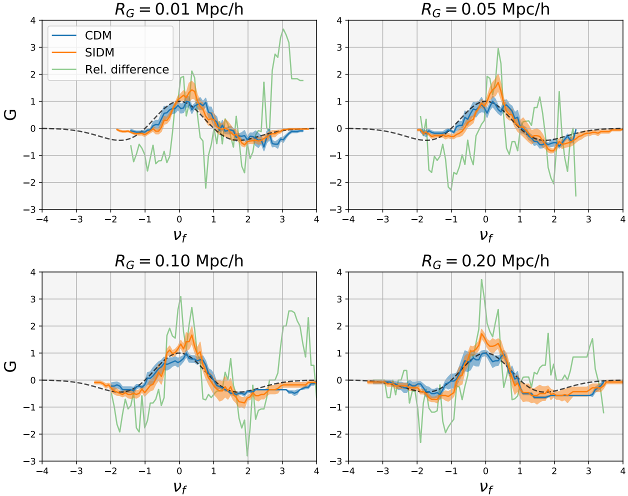

Figure: Genus computed on the halo density field.

Dependence on the redshift and cross-section

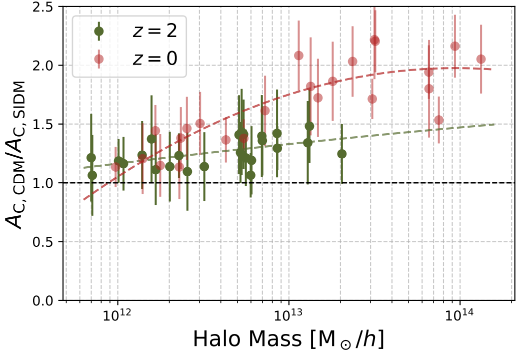

Figure: Smoothing scale $R_G = 10\,\mathrm{kpc}$. SIDM with $\sigma_0 = 1\mathrm{cm}^2/\mathrm{g}$.

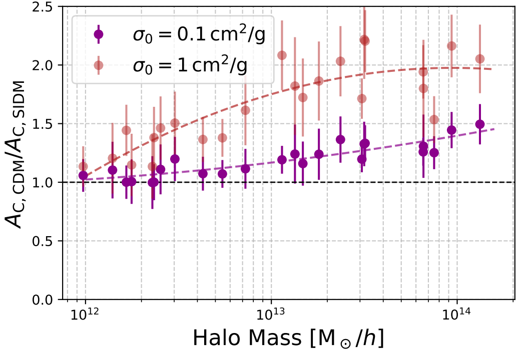

Figure: Smoothing scale $R_G = 10\,\mathrm{kpc}$ and $z = 0$.